

Counting Drones with Two LiDARs

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Abstract—We present a dataset for counting drones with two LiDAR sensors. Both sensors were calibrated against a motion capture system before recording.

I. INTRODUCTION

Small drones are hard to see in sparse point clouds. Our rig mounts a spinning sensor and a solid-state sensor side by side. The extrinsic calibration between them was estimated with GICP on ten stacked frames. This paragraph continues into the next column because the column ends here and

the sentence finishes at the top of the right one. Each recording lasts about two minutes.

A. Sensor Setup

The spinning sensor has 64 channels and a range of 120 m. The solid-state sensor reaches 450 m with a narrower field of view. Data were stored as rosbag files with hardware time synchronization.

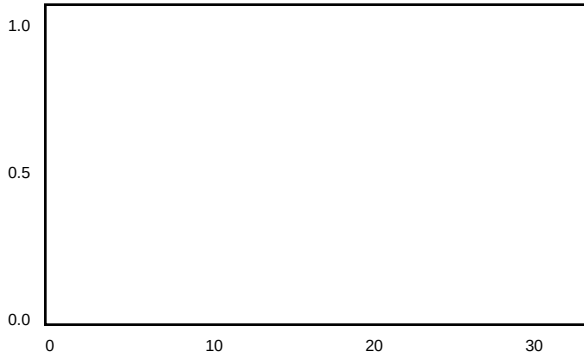


Fig. 1: Tracking error of both sensors over time.

II. RESULTS

The solid-state sensor tracked the drone in 100% of the sequences, the spinning sensor in 62%. Dense point clouds matter more than range for small targets.

REFERENCES

- [1] A. Example, “A survey of drone detection,” *Journal of Examples*, vol. 1, pp. 1–10, 2021.
- [2] B. Sample and C. Person, “Point cloud tracking,” in *Proc. Conference on Samples*, 2022.